We then bound terms $A_1(\theta)$, $A_2(\theta)$, and $A_3(\theta)$ in Eq. (A.3) seprately. For the term $A_1(\theta)$ in Eq. (A.3), we have

$$
\begin{aligned}
A_1(\theta) &:= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \Big[(w_{21}^V)^\top \cdot \sum_{i=1}^{M_{\text{train}}} \delta_{\tau,i}\delta_{\tau,i}^\top \cdot W_{11}^{KQ} x_{\tau,q}\Big]^2 \\
&\le \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \underbrace{\Big[\sum_{i=1}^{M_{\text{train}}} [(w_{21}^V)^\top \delta_{\tau,i}]^2 \cdot \sum_{i=1}^{M_{\text{train}}} [\delta_{\tau,i}^\top W_{11}^{KQ} x_{\tau,q}]^2\Big]}_{\text{by Cauchy-Schwarz Inequality}} \\
&\le \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \Big[\sum_{i=1}^{M_{\text{train}}} \max_{\|\delta_{\tau,i}\|_2\le\epsilon} [(w_{21}^V)^\top \delta_{\tau,i}]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \max_{\|\delta_{\tau,i}\|_2\le\epsilon} [\delta_i^\top W_{11}^{KQ} x_{\tau,q}]^2\Big] \\
&= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \Big[\sum_{i=1}^{M_{\text{train}}} [\|w_{21}^V\|_2 \cdot \epsilon]^2 \cdot \sum_{i=1}^{M_{\text{train}}} [\|W_{11}^{KQ} x_{\tau,q}\|_2 \cdot \epsilon]^2\Big] \\
&= \frac{2\epsilon^4 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \mathop{\mathbb{E}}_{\tau} \|W_{11}^{KQ} x_{\tau,q}\|_2^2.
\end{aligned} \tag{A.4}
$$

For the term $A_2(\theta)$ in Eq. (A.3), we have

$$
\begin{aligned}
A_2(\theta) &:= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \Big[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \cdot \sum_{i=1}^{M_{\text{train}}} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix} \delta_{\tau,i}^\top \cdot W_{11}^{KQ} x_{\tau,q}\Big]^2 \\
&\le \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \underbrace{\Big[\sum_{i=1}^{M_{\text{train}}} \Big[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}\Big]^2 \cdot \sum_{i=1}^{M_{\text{train}}} [\delta_{\tau,i}^\top W_{11}^{KQ} x_{\tau,q}]^2\Big]}_{\text{by Cauchy-Schwarz Inequality}} \\
&= \frac{2}{(N+M_{\text{train}})^2} \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} \Big[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}\Big]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} \Big[\max_{\|\delta_{\tau,i}\|_2\le\epsilon} [\delta_{\tau,i}^\top W_{11}^{KQ} x_{\tau,q}]^2\Big] \\
&= \frac{2}{(N+M_{\text{train}})^2} \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} \Big[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}\Big]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} [\|W_{11}^{KQ} x_{\tau,q}\|_2 \cdot \epsilon]^2 \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \|W_{11}^{KQ} x_{\tau,q}\|_2^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} \Big[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}\Big]^2.
\end{aligned} \tag{A.5}
$$

For the term $A_3(\theta)$ in Eq. (A.3), we have

$$
\begin{aligned}
A_3(\theta) &:= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \Big[(w_{21}^V)^\top \cdot \sum_{i=1}^{M_{\text{train}}} \delta_{\tau,i} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}^\top \cdot \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top\end{pmatrix} x_{\tau,q}\Big]^2 \\
&\le \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \underbrace{\Big[\sum_{i=1}^{M_{\text{train}}} [(w_{21}^V)^\top \delta_{\tau,i}]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top\end{pmatrix} x_{\tau,q}\Big]^2\Big]}_{\text{by Cauchy-Schwarz Inequality}} \\
&= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \Big[\sum_{i=1}^{M_{\text{train}}} \max_{\|\delta_{\tau,i}\|_2\le\epsilon} [(w_{21}^V)^\top \delta_{\tau,i}]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top\end{pmatrix} x_{\tau,q}\Big]^2\Big] \\
&= \frac{2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \Big[\sum_{i=1}^{M_{\text{train}}} [\|w_{21}^V\|_2 \cdot \epsilon]^2 \cdot \sum_{i=1}^{M_{\text{train}}} \Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top\end{pmatrix} x_{\tau,q}\Big]^2\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau} \Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}}\end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top\end{pmatrix} x_{\tau,q}\Big]^2.
\end{aligned} \tag{A.6}
$$

As a result, by inserting Eqs. (A.4), (A.5), and (A.6) into Eq. (A.3), we finally have that

$$\begin{aligned}
\mathcal{L}^{\text{adv}}(\theta) \leq\ & 2 \cdot \mathop{\mathbb{E}}_{\tau}\Bigg[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \cdot \frac{\begin{pmatrix} X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q} \\ Y_\tau & Y_\tau^{\text{sfx}} & 0 \end{pmatrix}\begin{pmatrix} X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q} \\ Y_\tau & Y_\tau^{\text{sfx}} & 0 \end{pmatrix}^\top}{N + M_{\text{train}}} \cdot \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} \cdot x_{\tau,q} - y_{\tau,q}\Bigg]^2 \\
& + \frac{2\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \mathop{\mathbb{E}}_{\tau} \|W_{11}^{KQ} x_{\tau,q}\|_2^2 \\
& + \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau} \|W_{11}^{KQ} x_{\tau,q}\|_2^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau}\Bigg[\begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}\Bigg]^2 \\
& + \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \sum_{i=1}^{M_{\text{train}}} \mathop{\mathbb{E}}_{\tau}\Bigg[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} x_{\tau,q}\Bigg]^2. \qquad \text{(A.7)}
\end{aligned}$$

The right-hand-side of Eq. (A.7) is exactly the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (9), which thus completes the proof. □

A.3 Proof of Theorem 1

This section presents the proof of Theorem 1, which is inspired by that in [69]. Specifically:

1. we first prove that terms w_{21}^V and w_{21}^{KQ} stay zero during the surrogate AT (Lemma A.6) via continuous gradient-flow, which thus can simplify the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ defined in Eq. (9) (Lemma A.7).
2. We then calculate a closed-form solution θ_* for the surrogate AT problem based on the simplified $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ (Lemma A.8), which is exactly the solution given in Theorem 1.
3. Finally, we prove that under the continuous gradient flow, the LSA model starts from the initial point defined in Assumption 1 can indeed converge to the closed-form solution θ_* (Lemma A.12), which thus completes the proof of Theorem 1.

We now start to prove the following Lemma A.6.

Lemma A.6. *Suppose Assumption 1 holds and the LSA model $f_{\text{LSA},\theta}$ is trained via minimizing surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (9) with continuous gradient flow. Then, for any continuous training time $t \geq 0$, we uniformly have that $w_{21}^V(t) = w_{21}^{KQ}(t) = 0_{d\times 1}$.*

Proof. When the LSA model $f_{\text{LSA},\theta}$ is trained with continuous gradient-flow, the updates of w_{21}^V and w_{21}^{KQ} with respect to the continuous training time $t \geq 0$ are given by

$$\begin{aligned}
\partial_t w_{21}^V(t) &:= -\partial_{w_{21}^V} \tilde{\mathcal{L}}^{\text{adv}}(\theta), \\
\partial_t w_{21}^{KQ}(t) &:= -\partial_{w_{21}^{KQ}} \tilde{\mathcal{L}}^{\text{adv}}(\theta).
\end{aligned}$$

Meanwhile, since Assumption 1 assumes that $w_{21}^V(0) = W_{21}^{KQ}(0) = 0_{d\times 1}$, therefore, to complete the proof, we only need to show that $\partial_t w_{21}^V(t) = \partial_t W_{21}^{KQ}(t) = 0_{1\times d}$ as long as $w_{21}^V(t) = W_{21}^{KQ}(t) = 0_{d\times 1}$ for any $t \geq 0$. In other words, below we need to show that $w_{21}^V = W_{21}^{KQ} = 0_{d\times 1}$ indicates $\partial_{w_{21}^V} \tilde{\mathcal{L}}^{\text{adv}}(\theta) = \partial_{w_{21}^{KQ}} \tilde{\mathcal{L}}^{\text{adv}}(\theta) = 0_{1\times d}$.

Toward this end, we adopt the notation in Eq. (9) to decompose the surrogate AT loss $\tilde{\mathcal{L}}(\theta)$ as follows,

$$\tilde{\mathcal{L}}^{\text{adv}}(\theta) := [\ell_1(\theta) + \ell_2(\theta) + \ell_3(\theta) + \ell_4(\theta)],$$

where

$$\ell_1(\theta) = 2\,\mathbb{E}_{\tau}\Big[((w_{21}^V)^\top \;\; w_{22}^V)\frac{\begin{pmatrix} X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q} \\ Y_\tau & Y_\tau^{\text{sfx}} & 0 \end{pmatrix}\begin{pmatrix} X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q} \\ Y_\tau & Y_\tau^{\text{sfx}} & 0 \end{pmatrix}^\top}{N + M_{\text{train}}}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} x_{\tau,q} - y_{\tau,q}\Big]^2, \tag{A.8}$$

$$\ell_2(\theta) = \frac{2\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2}\|w_{21}^V\|_2^2\,\mathbb{E}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2\Big], \tag{A.9}$$

$$\ell_3(\theta) = \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2}\,\mathbb{E}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2 \cdot \|((w_{21}^V)^\top \;\; w_{22}^V)\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}\|_2^2\Big], \tag{A.10}$$

$$\ell_4(\theta) = \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2}\|w_{21}^V\|_2^2 \cdot \mathbb{E}_{\tau}\Big[\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} x_{\tau,q}\|_2^2\Big]. \tag{A.11}$$

In the remaining of this proof, we will show that when $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$ holds, one has: (1) $\partial_{W_{21}^V}\ell_1(\theta) = \partial_{W_{21}^{KQ}}\ell_1(\theta) = 0_{1\times d}$, (2) $\partial_{W_{21}^V}\ell_2(\theta) = \partial_{W_{21}^{KQ}}\ell_2(\theta) = 0_{1\times d}$, (3) $\partial_{W_{21}^V}\ell_3(\theta) = \partial_{W_{21}^{KQ}}\ell_3(\theta) = 0_{1\times d}$, and (4) $\partial_{W_{21}^V}\ell_4(\theta) = \partial_{W_{21}^{KQ}}\ell_4(\theta) = 0_{1\times d}$, which thus automatically indicates that $\partial_{W_{21}^V}\tilde{\mathcal{L}}^{\text{adv}}(\theta) = \partial_{W_{21}^{KQ}}\tilde{\mathcal{L}}^{\text{adv}}(\theta) = 0_{1\times d}$.

Step 1: Show that $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$ indicates $\partial_{W_{21}^V}\ell_1(\theta) = \partial_{W_{21}^{KQ}}\ell_1(\theta) = 0_{1\times d}$. Such a claim can be directly obtained from the proofs in [69]. Specifically, when setting the (original) ICL prompt length from N to $(N + M_{\text{train}})$, the ICL training loss L in [69] is equivalent to our $\ell_1(\theta)$ defined in Eq. (A.8). Therefore, one can then follow the same procedures as those in the proof of Lemma 5.2 in [69] to show that the continuous gradient flows of W_{21}^V and W_{21}^{KQ} are zero when Assumption 1 holds. Please refer accordingly for details.

Step 2: Show that $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$ indicates $\partial_{w_{21}^V}\ell_2(\theta) = \partial_{w_{21}^{KQ}}\ell_2(\theta) = 0_{1\times d}$. Since the term w_{21}^{KQ} does not exist in the expression of $\ell_2(\theta)$ in Eq. (A.9), we directly have that $\partial_{w_{21}^{KQ}}\ell_2(\theta) = 0_{1\times d}$. Besides, for the derivative $\partial_{w_{21}^V}\ell_2(\theta)$, based on Eq. (A.9) we further have that

$$\begin{aligned}
&\partial_{w_{21}^V}\ell_2(\theta)\Big|_{w_{21}^V = 0_{d\times 1}} = \partial_{w_{21}^V}\Big[\frac{2\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \mathbb{E}_{\tau}\|W_{11}^{KQ}x_{\tau,q}\|_2^2\Big]\Big|_{w_{21}^V = 0_{d\times 1}} \\
&= \Big[\frac{4\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2} \cdot \mathbb{E}_{\tau}\|W_{11}^{KQ}x_{\tau,q}\|_2^2 \cdot (w_{21}^V)^\top\Big]\Big|_{w_{21}^V = 0_{d\times 1}} \\
&= \frac{4\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2} \cdot \mathbb{E}_{\tau}\|W_{11}^{KQ}x_{\tau,q}\|_2^2 \cdot 0_{d\times 1}^\top = 0_{1\times d},
\end{aligned}$$

which justifies our claim in Step 2.

Step 3: Show that $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$ indicates $\partial_{w_{21}^V}\ell_3(\theta) = \partial_{w_{21}^{KQ}}\ell_3(\theta) = 0_{1\times d}$. We first rewrite $\ell_3(\theta)$ that defined in Eq. (A.10) as follows,

$$\begin{aligned}
\ell_3(\theta) &= \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2}\,\mathbb{E}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2 \cdot \|((w_{21}^V)^\top \;\; w_{22}^V)\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}\|_2^2\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \cdot \mathbb{E}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2\Big] \cdot \sum_{i=1}^{M_{\text{train}}}\mathbb{E}_{\tau}\Big[((w_{21}^V)^\top \quad w_{22}^V) \cdot \begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}^\top \cdot ((w_{21}^V)^\top \quad w_{22}^V)^\top\Big] \\
&= \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \cdot \mathbb{E}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2\Big] \cdot ((w_{21}^V)^\top \quad w_{22}^V) \cdot \left(\sum_{i=1}^{M_{\text{train}}}\mathbb{E}_{\tau}\Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}^\top\Big]\right) \cdot ((w_{21}^V)^\top \quad w_{22}^V)^\top.
\end{aligned} \tag{A.12}$$